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Unit :- 04

Q. 1 Define Gross calorific value and Net calorific value & give relation between them. What are the three main classification of fuels ? Give example.

Ans:- Gross Calorific value :-

- It is amount of heat produced when unit quantity of fuel is burnt compeletely in presence of oxygen and the products of combustion are cooled at room tempe.

Net Calorific value :-

- Net calorific value is defined as amount of heat obtained practically on complete combustion of unit mass of solid liquid or gas fuel at STP after skipping all product of combustion.

Relation :-

$$GCV = NCV + \left(\frac{9 \times h \times \text{Latent heat of water}}{100} \right)$$

where,

h % = % of hydrogen present in fuel

latent heat of water = 587 cal/gm
= 587 Kcal/kg
= 2450 Kj/kg

- classification of fuels :-
- On Basis of occurance :-
- primary fuel :-
- The fuel which occur in nature and is used directly as fuel ex. wood, coal, dung.
- secondary fuel :-
- The fuel which are derived or prepared from primary. Ex. petrol, Bio-diesel.

- on Basis of availability :-
- Renewable fuels :-
- These are energy sources which does not vanish

or which can be regenerated many times.
Ex. sunlight

- Non-Renewable fuels :-
- These are energy resources which vanishes with time or which can not be regenerated.
Ex. Coal.

- on Basis of state of matter :-
- solid fuel :- The fuel which is in solid state is called solid fuel.
- Ex. dung, wood
- liquid fuel :- The fuel which is in liquid state is called liquid fuel.
- Ex. petrol, diseal.
- Gas fuel :- The fuel which is in gaseous state is called gas fuel.
- Ex. LPG, CNG., H_2.

Q.2 Explain any five characteristics of ideal fuel.

Ans:- ① High calorific value :-
- calorific value is amount of heat produced by complete combustion of fuel.
- If fuel has high calorific value then large quantity of Heat obtained by burning of small amount of fuel.
- solid fuel may be low CV, liq and gas fuel may have high CV.

② moderate Ignition point :-
- Ignition point is the temperature at which fuel starts its combustion.
- IP of fuel is very low then its easily catches fiere even at room temperature.
- If IP of fuel is very high it is difficult to start combustion
- So it always supposed as ideal fuel if any fuel ~~an~~ may have moderate IP.
- Gaseous fuel may very low IP. solid fuel harer high IP. liquid fuel is moderat IP.

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③ low cost [cheap]
- Any fuel which is cheap any easily available called good fuel.
- liquid and gas fuel may costly but solid fuel is cheap.

④ small space to store :-
- fuel which can be stored in very small space called good fuel
Generally gas fuel requires large space to store.
- liquid fuel requires moderate space to store and solid fuel requires small space to store.

⑤ Used in internal Combustion engine :-
- If any fuel used in internal combustion engine then called good fuel.
- liq. and gaseous fuel are easily used I.C. as fuel flow rate can be easily controlled.
- Solid fuel can't be controlled.

Q.3 A fuel containing 12% of hydrogen and 12% of moisture has GCV = 35000 Kj/kg. find NCV of fuel.

Given,
$h = 12\%$
$GCV = 35000$
moisture $= 12\%$
% of hydrogen in moisture,

$18 \rightarrow 2$
$12 \rightarrow ?$

$$x = \frac{24}{18} = 1.333\%$$

Now totalal,
$\%h = 12 + 1.33$
$= 13.33\%$

then,
we know

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$$GCV = NCV + \left(\frac{9 \times \% h \times \text{latent heat of water}}{100} \right)$$

$$NCV = 35000 - \frac{9 \times 13.33 \times 2450}{100}$$

$$= 35000 - 2939.265$$

$$NCV = 32060.735 \ kJ/kg$$

Q4 Define the terms:-

1] Fuel :-

- Fuel is a substance which combustion with ~~oxygy~~ oxygen gives energy / heat for domestic or indistrial use.

2] Gross Calorific value :-

- Gross calorific value is the amount of heat/energy produced from unit quantity of fuel in presence of oxygen and combustion productors are cooled at room temperature.

3] Biodiseal :-

- It is a mixture of methyl esters and long chain carboxylic acids.

Q.5 The Coal sample containing 7% hydrogen (dry) has net Calorific value of 3990 Kcal/kg. Calculate the Gross Calorific value of Coal if latent heat of water vapor is 585 cal/g.

- Given,

$H = 7\%$

$NCV = 3990 \ Kcal/kg$

$GCV = ?$

latent heat = 585 Cal/g

Then, we know,

$$GCV = NCV + \left(\frac{9 \times \% h \times \text{latent heat of } H_2O}{100} \right)$$

$$GCV = 3990 + \left(\frac{9 \times 7 \times 585}{100} \right)$$

$$= 3996 + 368.55$$

$$GCV = 4358.55$$

Q.6 Define power alcohol. Explain ~~in~~ reactions involved in preparation of power alcohol and give any four advantage and disadvantages.

- When ethyl alcohol is used as fuel in internal combustion engine then it is called power alcohol.
- Preparation :-

$$\underset{\text{starch}}{n(C_6H_{10}O_5)} + n/2\,H_2O \xrightarrow[\text{Enzyme}]{\text{Diastse}} \underset{\text{matose}}{\frac{n}{2}(C_{12}H_{22}O_{11})}$$

$$\underset{\text{maltose}}{\frac{n}{2}(C_{12}H_{22}O_{11}} \xrightarrow{\text{maltose}} \underset{\text{glucose}}{n(C_6H_{12}O_6)}$$

$$C_{12}H_{22}O_{11} \xrightarrow{\text{Invertose}} n(C_6H_{12}O_6) + (C_6H_{12}O_6)$$

$$\underset{\text{Glucose / fructose.}}{(C_6H_{12}O_6)} \xrightarrow{\text{zymase}} \underset{\text{ethyl alcohol}}{2C_2H_5OH} + 2CO_2$$

- Advantages :-
- It is cheaper than petrol
- large octane number 90.
- Reduces dependency on other contry.
- Alcohol has property to sock the moisture present in petrol.

- Disadvantages
- It has low calorific value (7000 Ca/gm).
- It reduces power output.
- Power alcohol undergoes oxidation and harms.
- Engine modification must be done before use of alcohol.

Q.7 Define Biodiseal explain preparation of biodiseal and given any four advantages and disadvantages.

- It is the mixture of methyl ester of long carboxylic acids.
- Preparation:-

$$\begin{array}{l} CH_2-OOC-R_1 \\ | \\ CH-OOC-R_2 \\ | \\ CH_2-OOC-R_3 \end{array} + 3\,CH_3-OH \xrightarrow[\text{Na-m}]{CH_3ONa} \begin{array}{l} CH_3-COO-R_1 \\ + \\ CH_3-OOC-R_2 \\ + \\ CH_3-OOC-R_3 \end{array} + \begin{array}{l} CH_2-OH \\ | \\ CH_2-OH \\ | \\ CH_2-OH \end{array}$$

- while preparing biodiseal from vegetable oil we get glysorol.
- This glysorol can easily seperated from bio-diseal.
- Advantages:-
- It has high cetane number 45-55.
- Environmental and ecofriendly.
- It don't ~~proced~~ ~~proces~~ produces Co.
- It is clean energy.
- Disadvantages:-
- It can't be used in cold region.
- Cloud or pour point of biodiseal is high.
- It dissolves rubber in it.
- Biodiseal strongly adhere on metal.

Q.8 Explain process of production of hydrogen gas by steam reforming of methane and coke. Give two uses of hydrogen.

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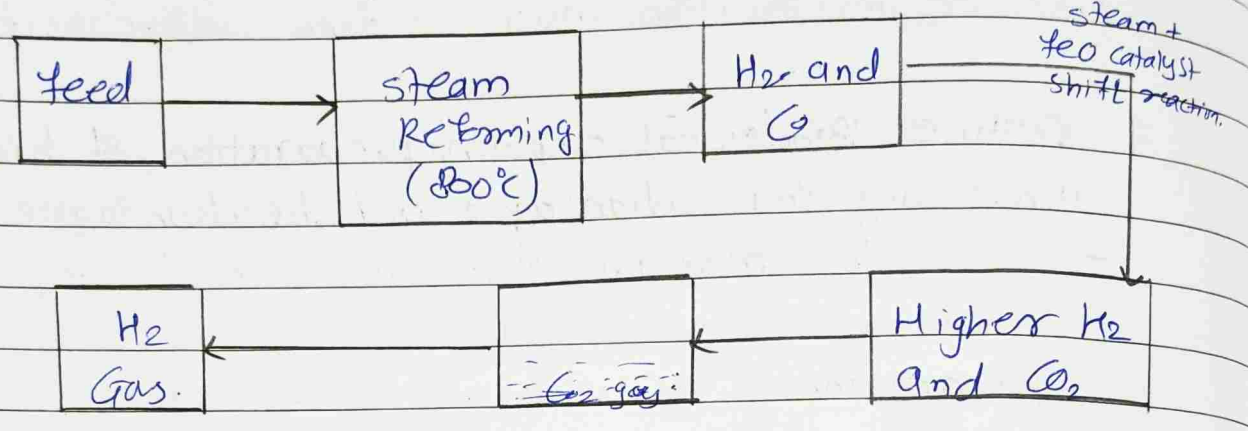
- production of hydrogen gas:- 1) from methane
- It is the process of reacting steam with hydrogen in the presence of catalyst at high temperature to obtain hydrogen and CO_2.



- Reaction:-

$$CH_4 + H_2O \xrightarrow{Ni/feO\ 800°C} CO + 3H_2$$

- Shift Reaction:-

$$CO + H_2O \xrightarrow{feO\ 370°C} CO_2 + H_2.$$

2) from Coal

Reaction:-

$$C + H_2O \xrightarrow{700°C} CO + H_2$$

Shift ~~reaint~~ reaction

$$CO + H_2O \xrightarrow{feO\ 370°C} CO_2 + H_2$$

- Disadvantages:-
- It is very hard to liquify and store.
- difficult to cool & for liquify.
- to relese the hydrogen from the metal requires high energy.
- to store the hydrogen gas in metal requires more time.

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Advantages :-
- It is used as fuel in engines.
- It has low polluting ability.
- H_2 gas is ~~bio~~ ecofriendly.
- In future the hydrogen gas in future fuel.

Q9 Explain in details importance of hydrogen gas as a future fuel. State the difficulties in the storage and transportation of fuel.
- It is going to be 21st century fuel.
- It is going to be used widely in domestic industrial and transportation fuel.
- It produces very high temperature and produce no pollution it is called as zero emmision fuel.
- After vanishing crude oil the Hydrogen gas as a fuel will be used widely in domestic or industrial use as a energy source.
- Diffulties in storage or transportation.
- It is highly inflammble as ignition point is lowest.
- It is lightest gas from any other gas so it will stored very in less amount.
- It is very dangrous to store or refilling. it will be explode even a small amount of air comes in contact.
- It is diffuclt to store and liquity.

Q. 10 Give principle, construction and working of lithuim ion battery with neat labbled diagrams and advantages.
- By movment of electron through outer circuit and Li ions through electrode between cathode and anode Li ion battery converts chemical energy to electrical energy and electrical energy to chemical energy.

- Construction :-

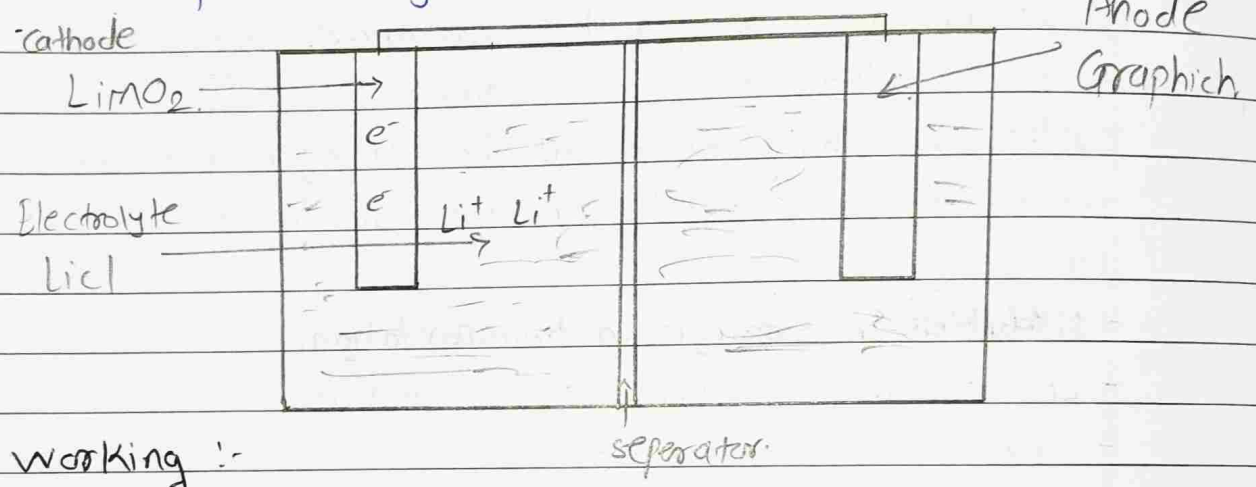
Anode :- Typically made of graphite, where lithium ions are stored during charging.

Cathode :- Typically made of lithium compound (e.g. $LiCoO_2$, $LiMn_2O_4$), where lithium ions are stored during discharging.

Electrolyte :- A liquid or gel containing a lithium salt (e.g $LiPF_6$) in an organic solvent, which allows lithium ions to move between the electrode.

seperator :- A thin, porous membrane that physically seperates the anode and cathode while allowing lithium ions to pass through. (while charging).

- Cathode



- Working :-
- While Charging At cathode :-
- on external power source electrons are generated and Li^+ Ions are generated at Cathodes.
- Reaction :-

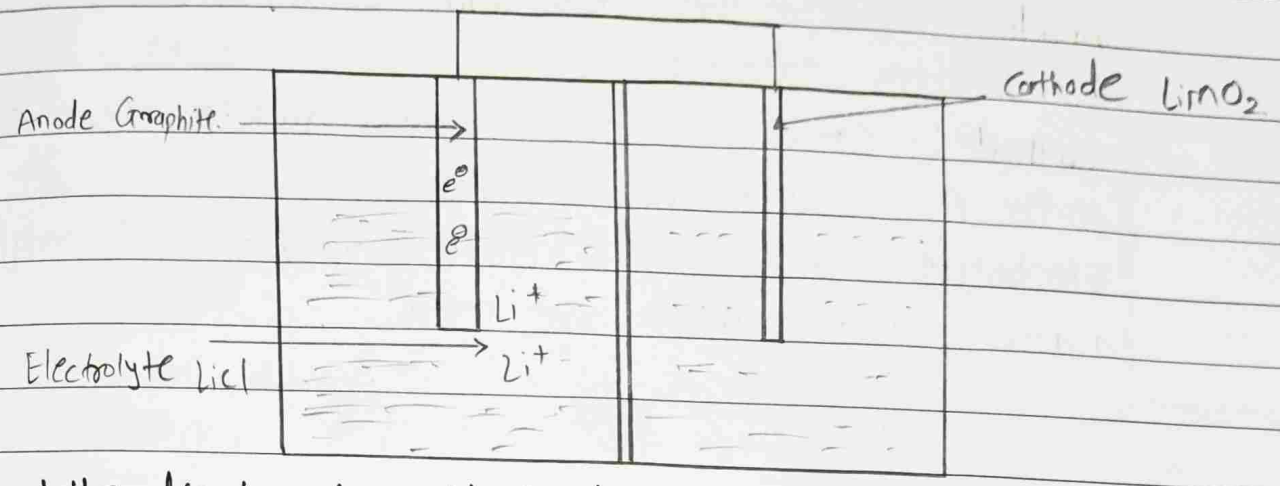
$$LiMO_2 \rightarrow Li^+ + MO_2 + e^{\ominus} \quad (oxd^n)$$

- The electron will move from cathode to anod through outer ckt.
- also Li ion moves from Cathode to anode through electrolyte and seperator.
- Reaction at anode :-

$$C_6 + Li^+ + e^{\ominus} \rightarrow LiC_6$$

- At anode Li ion and graphite forms lithium graphite.

(Inwhile discharging)



- while discharging At Anode:-
- Reaction:-

$$LiC_6 \rightarrow 6 + Li^+ + e^{\ominus} \quad (Oxd^n)$$

- At Anode:-

$$Li^+ + MO_2 + e^{\ominus} \longrightarrow LiMO_2 \quad (Red^n).$$

- The electrons will move from Anode to cathode through outer circuit Lions will move from anode to cathode through electrolyte and seperator
- Advantages:-
- High energy density
- Lightweight and Comfortable.
- fast charging.
- No memory effect.
- longer life time.

Q.11 State the principle, Construction and working of H_2O_2 fuel cell with neat lablled diagram. Give the advantage.

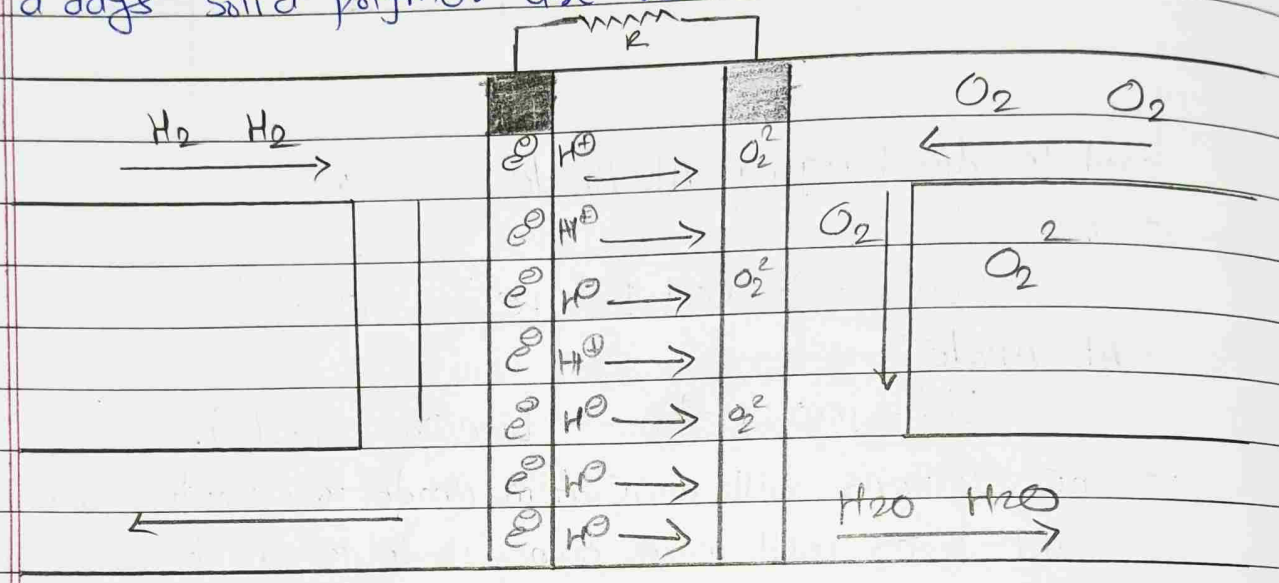
- principle:- fuel cells is the device which fuel get oxidized at anode by the oxidant O_2 from cathode converting chemical energy into electricity.

Construction :-

Anode :- anode is the porous graphite electrode coated with pt powder as catalyst.

Cathode :- Cathode is the porous graphite electrode coated with Ag powder as catalyst.

Electrolyte :- 30% - 85% KOH is used as electrolyte now a days solid polymer are used.



Working :-

- when H_2 is passed through anode, the pt catalyst does oxid^n of H_2 and H_2 becomes H^+.
- Reaction

$$2H_2 \longrightarrow 4H^+ + 4e^{\ominus}$$

- Then H^+ ions moves from anode to cathode through electrolyte
- meanwhile the $e^{\ominus}$ moves from anode to cathode through outer ckt.
- At the same time at cathode an O_2 get reduced in pr. of silver and forms O^{2-}
- this O_2^- and from anode an H^+ are coming in large amount they both get reacts with each other and water and electricity is formed.
- Reaction :-

$$O_2 + 4e^{\ominus} \longrightarrow O_2^{2-} + \text{energy}$$

$$4H^+ + 2O^{2-} \longrightarrow H_2O$$

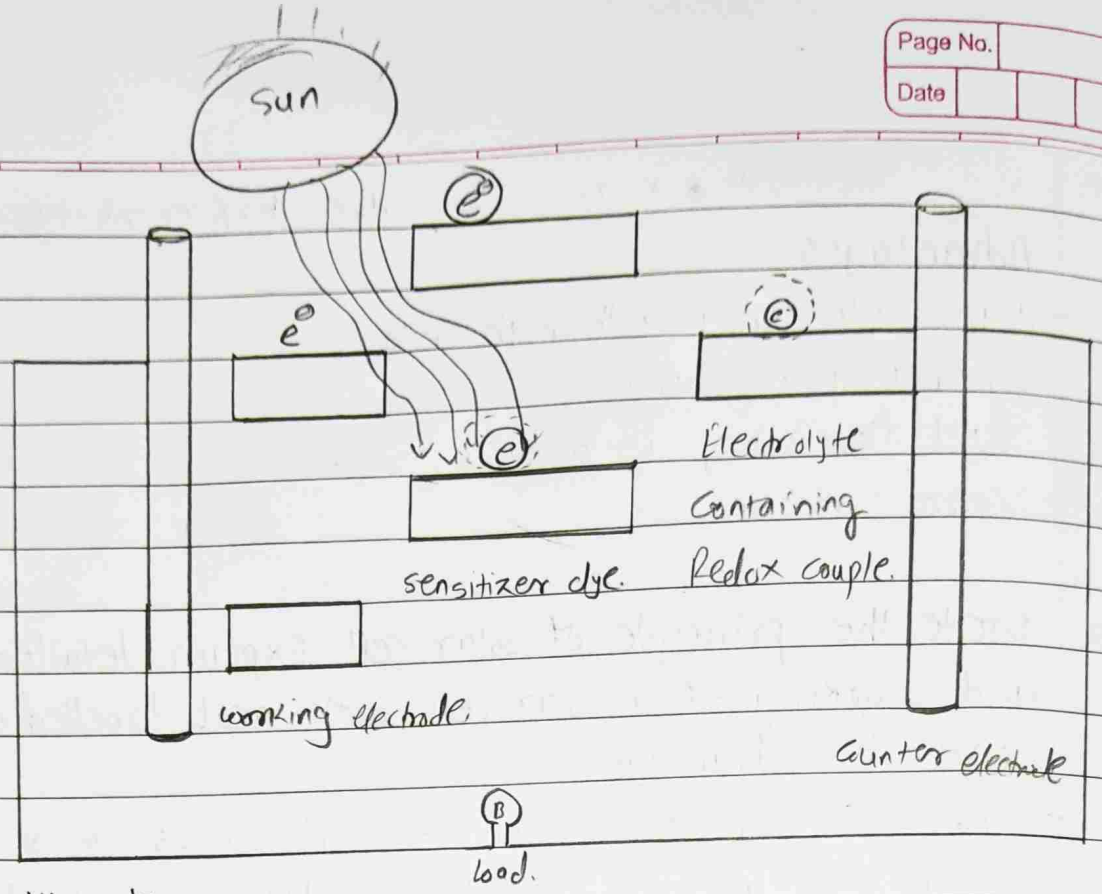
$$2H_2 + O_2 \longrightarrow 2H_2O + \text{Electricity}$$

- Advantages:-
- High effeciency with high energy.
- Continous powe source
- fuel flexibility is more.
- Zero emission.

Q.12 State the principle of solar cell. Explain detailed construction and working of a solar cell with neat labelled diagram. Give its advantages.

- principle :- In a solar cell, sunlight excites electrons in the dye which are injected into the conduction band of a semiconductor. The electrons flow through the external circuit to produce electricity and are regenerated by electrolyte.
- Construction :-

- Working Electrode (Anode) :- the working electrode of Dye sensitized solar cell is coated with wide band gap semiconductors like TiO_2 or ZnO nanoparticles.
- Sensitizer :- It is the part of a DSSC whose purpose is to absorb the maximum light to excite more and more numbers of electrons from ground state to exicted state.
- Electrolyte :- It is the substace which fills the space betn anode and cathode and which helps to transfer the ele$^{\ominus}$ from anode to cathode or vice versa.
- Counter electrode :- The other end is filled with counter electrode that catches the el$^{\ominus}$ send from the anode basically it acts as collector. It is made up of platinum.



Working :-

- when sunlight falls on dye the electrons present on dye comes in exicted state due to exited state they jump ~~from~~ to conduction band ~~to~~ from valence band.
- After some time the comes to normal and try to jump valence band but the wide band gap catches electrons while they jump in valence band.
- once the electrons get captured by wide band gap the electrons can't go to valence band or exited state.
- also there is formation of holes and electrons also they both can't get recombines.
- Now the electron flow to the cathode from anode
- ~~the~~ and the electricity generated ~~th~~
- The electrons comes from ckt to counter electrode that electrode combines the hole and electron and this process get repeated until the sunlight falling on dye.

- Advantages :-
- Clean Energy produced from Solar cell.
- Low water usage. In this process.
- Reduces electricity bills.
- This is reliable.